

Nicholas D. Gray, Ph.D.

RESEARCH SCIENTIST

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Research scientist with a Ph.D. in Cognitive Psychology, nearly four years of insurance analytics, and nine years designing and analyzing survey and experimental studies. Works daily in Python and SQL over large, complex datasets — cleaning, modeling, validating, and automating the reporting around them — on a foundation of measurement design and statistical inference from ANOVA through structural equation modeling.

TECHNICAL SKILLS

Programming & Data	Python (proficient), SQL / Microsoft SQL Server, R, SPSS, MATLAB, Microsoft Excel (proficient), ETL & data cleaning
Statistical Methods	Descriptive & inferential statistics, ANOVA, Regression (linear & logistic), Multivariate analysis, Structural equation modeling, Meta-analysis
Modeling & Machine Learning	XGBoost & gradient boosting (applied), Model validation & evaluation, Automated analysis pipelines, LLM-assisted reporting
Survey & Study Design	Survey & instrument development, Qualtrics, Sampling & power analysis, Experimental design, Qualitative interviews & analysis, Mixed-methods integration
Communication	Stakeholder reporting & dashboards, Data visualization, Technical writing, Manuscript writing & presentation

PROFESSIONAL EXPERIENCE

Data Scientist <i>Safety National Casualty Corporation · St. Louis, MO</i>	Oct 2022 – Present
- Extract and join data from large, complex sources in Microsoft SQL Server, then clean, transform, and model it in Python. - Run, adjust, and validate established machine-learning models (XGBoost), interpreting predictions for underwriting and claims questions. - Built an automated reporting pipeline that calls an LLM to generate written insights alongside the quantitative results. - Design custom analyses out of exploratory investigation and build the reports and dashboards that deliver them to internal teams and external stakeholders.	
Postdoctoral Researcher — Quantitative Analysis <i>Institute for Successful Longevity, Florida State University · Tallahassee, FL</i>	Aug 2019 – Oct 2022
- Led statistical analysis for large-scale behavioral clinical trials, working across longitudinal survey and assessment datasets and multiple analysis platforms. - Designed and fielded survey instruments in Qualtrics, and established statistical performance norms for the measures collected. - Built regression models identifying the variables that drive participant outcomes, and integrated qualitative interview data with quantitative measures. - Published results in peer-reviewed journals and presented findings at national conferences.	

Graduate Researcher

Aug 2013 – May 2019

Department of Psychology, Florida State University · Tallahassee, FL

- Modeled experimental data with logistic regression and related statistical methods across several software platforms.
- Designed, programmed, and ran grant-funded experimental research; managed the lab, recruiting and supervising undergraduate researchers.
- Taught laboratory sections of Cognitive Psychology for three years and co-instructed an online Careers in Psychology course, translating research concepts and methods for non-expert audiences.

SELECTED PROJECTS

adherence – Python · github.com/ndgl3b/adherence

Scores how regular a behavior is — not just how often it happens — from nothing but a list of timestamps, then forecasts when the next session will land. The same model applies to medication timing, exercise, physiotherapy, or a training program.

CivicGateway – JavaScript · civicgateway.org

A nonpartisan site showing upcoming elections and elected officials by city; static front end over a Supabase/PostgreSQL back end with a schema designed from scratch, row-level security, and an offline fallback.

tuned-in – JavaScript · ndgl3b.github.io/tuned-in

A teaching collection pairing pop-culture clips with the cognitive psychology they demonstrate — and, just as often, get wrong. Each concept is labeled accurate, overstated, or myth.

EDUCATION

Ph.D., Cognitive Psychology

2019

Florida State University

Thesis: The causes of and influences on being reminded

M.S., Cognitive Psychology

2016

Florida State University

Thesis: Avoiding interference: Differentiation and reminding

B.S., Psychology, cum laude

2012

Ohio State University

Certification: Machine Learning Specialization — DeepLearning.AI & Stanford Online (Coursera), 2026. Supervised learning (regression, classification, neural networks, decision trees), unsupervised learning, recommender systems, reinforcement learning

Graduate methods training: ANOVA, regression, meta-analysis, structural equation modeling

Additional coursework: Machine Learning with Python; Microsoft SQL Server (independently completed)

SELECTED PUBLICATIONS

Gray, N., & Charness, N. (2022). Technology obsolescence across the adult lifespan in a USA internet sample. *Frontiers in Public Health*, 10, 1005822.

Gray, N., Yoon, J.-S., Charness, N., Boot, W. R., et al. (2021). Relative effectiveness of general versus specific cognitive training for aging adults. *Psychology and Aging*, 37(2), 210–221.

Carr, D. C., Tian, S., He, Z., et al., incl. Gray, N. (2022). Motivation to engage in aging research: Are there typologies and predictors? *The Gerontologist*, gnac035.

Charness, N., Boot, W. R., & Gray, N. (2020). Mobile monitoring and intervention technology (MMI) for adaptive aging. In *Mobile Technology for Adaptive Aging: Proceedings of a Workshop* (pp. 21–40). National Academies Press.

Conference presentations at the Psychonomic Society (2018) and the Cognitive Aging Conference (2022).